CRYPTONITE

RESEARCH TASK PHASE

Module-3: Clustering and Dimensionality Reduction

Core Topics

- k-Means Clustering and DBSCAN (Density-Based Clustering)
- Principal Component Analysis (PCA)
- Support Vector Machines (SVM)

Dataset

- Football Players Stats (2025–2026)

Task List

1. **Clustering:**
   - Implement either **K-Means** or **DBSCAN** from scratch on the given dataset. Structure your implementation using appropriate **classes and functions**.
   - Include visualizations that help explain and analyze your clustering results.
   - Implement either the **Elbow Method** or **Silhouette Score** from scratch to evaluate your clustering approach.
   - Clearly justify your **data preprocessing steps** and **choice of features**.
   - Provide a comprehensive analysis of the clusters with respect to the domain of the dataset. Interpret what the identified clusters represent in terms of football players and their characteristics.
2. **PCA and SVM:**
   - Prepare a short written report of approximately **1–2 pages** explaining **Support Vector Machines (SVM)** and **Principal Component Analysis (PCA)**.
   - Briefly cover the mathematical intuition behind both techniques, what they are, why they are used, and their major advantages and limitations.
   - Implement **PCA from scratch** on the dataset.
   - Visualize your clusters in a reduced feature space of **2 or 3 dimensions**.
3. **Bonus Task:** Read and understand the paper:
   - Data Clustering: 50 Years Beyond K-Means

Basic Resources

- **Introduction to PCA:** Article, Video
- **K-Means Clustering:** Article, Video
- **DBSCAN Clustering:** Article, Video
- **Support Vector Machines:** Article
- **Coursera (Deeper Coverage):** Machine Learning with Python (Manipal/Coursera)

Evaluation and Submission Guidelines

- You already have private GitHub repositories. Create a folder named: `Task-3`

  The folder should contain:

  - Jupyter notebooks containing your implementations and analysis.
  - A concise LaTeX report covering **SVM and PCA**.
  - A `Task-3/README.md` file containing a short summary of your approach, results, and key findings.

- Evaluation will primarily focus on:

  - Correctness of implementation
  - Conceptual understanding
  - Code quality and structure
  - Appropriate use and reporting of evaluation metrics
  - Quality of analysis and interpretation

Notes & Reminders

- Use explicit evaluation metrics wherever applicable and include meaningful visualizations.
- You may use Google Colab or Kaggle for computation if required.
- Contact the mentors if you have any queries:

  - Mohammed Luqmaan Akhtar (+91 97489 06653)
  - Vaibhav Chouraisa (+91 89104 30428)

- You will be thoroughly interviewed on all concepts covered in this task as well as your implementation and code. Make sure you understand everything in detail.
- **TASK-3 DEADLINE: 30th October, 2026**